

UCL SGNL OPS / Orbit Dynamics Library

Assessed from the working tree at ~/od120lite, x86-64 Linux, GCC 15

Verdict

As a sellable product, close to zero. As a validated reference implementation and a carrier of one genuinely scarce capability, worth keeping — but that capability's data is not in this tree. The first action is not engineering.

What I measured

Build state	11 of 11 smoke checks pass, x86-64 Linux / GCC 15
Integrator quality	RK8/7 agrees with the analytic two-body solution to 1.4 cm over one sidereal day at GEO, 60 s step
Throughput	0.81 s per one-day arc at 70×70 gravity + third body + solid and pole tide + GR, 10 s step; 0.07 s at 12×12
Size	15,107 LOC library (54 files), 4,764 LOC headers, 785 LOC across 7 CLI utilities, 13,863 LOC vendored
Authors	David Harrison (98 files), Santosh Bhattarai (38), Zhen Li, Stuart Grey, Alex Forsyth, Marek Ziebart — 2014–2017

The physics is real and the numerics are sound. That is not the problem.

What is in the box

Force models. Earth gravity to 360×360 across eight fields (EGM96, EGM2008, GGM03C, JGM-3, GRIM5-S1/C1, GGM01C/S) with a full analytic gradient; third body (Sun, Moon, eight planets) via DE405 or anything CALCEPH reads; solid Earth tide; pole tide; time-variable gravity; general-relativistic correction; drag (USSA76, TIE-GCM tables, NRLMSISE-00); solar radiation pressure (cannonball, box-and-wing, UCL grid-file); Earth radiation pressure (simple, and a CERES mesh at five resolutions); three thermal re-radiation models; antenna thrust; Y-bias; Lorentz force with IGRF-11/12.

Propagators. Keplerian, RK4, RK8/7, RKF7/8, SGP4.

Frames and data. J2000 $\leftrightarrow$ ECEF on IAU1980 nutation with IERS EOP now spanning 1962-01-01 to 2027-09-16; planetary ephemeris now 1978-12-14 to 2060-01-30.

Where the value actually sits

The UCL lineage is known for one thing: **high-fidelity non-conservative force modelling** — ray-traced solar radiation pressure and thermal re-radiation for GNSS spacecraft. The tree contains the two-layer solar-panel TRR model driven by commanded power density, the grid-file SRP interpolation engine, the CERES ERP mesh, and 27 curated optical and thermal property sets including GPS-IIR (with and without the nadir antenna panel), Galileo IOV and FOC, BeiDou IGSO and GEO, Sentinel-1A, Jason-2, SPOT-4, and HAMR/LAMR debris analogues.

That is scarce. Detailed per-satellite ray-traced SRP combined with panel thermal modelling is still the preserve of a handful of analysis centres.

The hole in the middle

There are zero .grd files in the tree. Every grid-file path in `src/Resident_constants.cpp` points at `../analyses/<mission>/*_gridfiles/`, and none of those directories exist. `analyses/` holds two student CubeSat studies and nothing else. The differentiated capability is an engine with no fuel. What can actually run today — cannonball and box-and-wing SRP — is commodity.

This is the single most important finding in this assessment.

What it cannot do

BLOCKING for any commercial use

- **No licence, no copyright headers, no git history.** Provenance is UCL; under standard UK university IP terms the code belongs to UCL, not to whoever holds the tree. Nothing can be sold or open-sourced until that is settled.
- **No observation models.** No pseudorange, carrier phase, radar range or Doppler, or optical angles. It fits to *already-determined* precise orbits in SP3/JPL format. It is a force-model validation tool, not an orbit determination system.
- **No covariance.** No covariance propagation, no process noise, no uncertainty quantification. That rules out conjunction assessment, space situational awareness, and anything an operator would call operational.

DEGRADING

- Estimation is **six-state only** — no SRP scale factor, drag coefficient or empirical accelerations, which is exactly what operational precise orbit determination estimates. The code comments sketch the sensitivity matrix $S = \partial y / \partial p$ and then do not implement it.
- **Drag and SRP contribute no partial derivatives** to the state transition matrix, and the STM is propagated in RK4 and RK7/8 but **not in the default RK8/7**.
- No manoeuvre modelling of any kind.
- `main_fit_orbit_to_sp3_v3.cpp` takes **no arguments** — the config and data paths are string literals in `main()`. Every new fit is a recompile.
- 1,175 lines of `if (name == "gpsIIR")` spacecraft definitions inside the library, directly contradicting the README's own @@@@ No spacecraft specific code in library @@@.
- Single-threaded, one object per process. Catalogue-scale work would need re-architecting.
- Global mutable statics doing relative-path file I/O at load time (`Flux_earth_ceres::m_ERPgrid` reads `../res/earth_radiation` during static initialisation). It is a CLI, not an embeddable library.
- No Python or C bindings. The market expects Python.
- IAU1980 frames rather than IERS 2010 / IAU2006-2000A. Defensible with observed celestial pole offsets, but a due-diligence red flag.
- One shell smoke test of 11 assertions, written during the revival. No unit tests, no CI, no regression suite against published benchmarks.
- Data currency is manual: IGRF-12 (IGRF-14 has been current since November 2024), CERES as a twelve-month climatology, EOP and space weather on shell scripts someone must remember to run.

COSMETIC

- `png++` is vendored and unused, yet the Makefile still links `libpng`.
- EGM2008 is wired in as gravity model 8 but undocumented in the config file.

What it is up against

Orekit (Apache-2.0, Java with Python bindings, full IERS 2010, real measurement types, covariance, manoeuvres, conjunction, extensive test suite, used operationally), GMAT, GODOT, Basilisk, Tudat and Nyx are all free and better governed in the commodity layer. Commercially the field is ODTK, FreeFlyer, COMSPOC and the SSA service providers. A 2017 single-threaded C++ CLI with no licence, no tests, no covariance and no measurement models does not compete with any of them on propagation.

Three ways to put a number on it

- **Sale value as software: approximately £0.** No licence to sell under, no differentiation in the commodity layer, no support story, and free competitors that are better.
- **Replacement cost: roughly £300k–£600k** at UK specialist rates for around three person-years of domain-expert work. This is not market value — you would not rebuild it, you would adopt Orekit.
- **Know-how and services: the only real line item.** UCL-lineage non-conservative force modelling sold as analysis services to GNSS operators, altimetry missions, or SSA firms needing better SRP and area-to-mass handling for high area-to-mass ratio debris. The Cone_HAMR and Large_Sphere_HAMR entries suggest that direction was already open.

Options, ranked

1. **Archive as a reference implementation.** Cost approximately zero. Keep it as a validation oracle and a teaching asset — it builds, it runs, the integrator is demonstrably good, and the revival work (Eigen swap, EOP merge, JPL-to-FECsoft converter, CALCEPH integration, NRLMSISE-00 drag, TIE-GCM bounds fix) makes it usable for student projects and papers. *This is the honest answer for the library as a whole.*
2. **Harvest the differentiated physics into a modern host.** Port the TRR panel model, grid-file SRP and CERES ERP into Orekit or a standalone C++/Python module, publish it, and build reputation and consultancy on it. Contingent on UCL IP clearance *and* on locating the grid files and the ray-tracer that generates them.
3. **Productise the whole thing. Not recommended.** Covariance, measurement models, manoeuvres, tests, parallelism, bindings, IERS 2010, licence — a multi-year rebuild that lands you where Orekit already is.

Recommendation: option 1 now; option 2 only if the three questions below come back favourably.

What to find out first — none of it is engineering

1. **Who owns this?** UCL's position on the 2014–2017 code, and whether anything can be licensed or released.
2. **Do the SRP grid files and, more importantly, the ray-tracer that generates them, still exist anywhere?** Without the generator you cannot model a new spacecraft, and the engine is worthless.
3. **Is there a named customer for GNSS-grade SRP and TRR analysis?** If yes, the asset is a services business. If no, it is an archive.

Basis: measured facts come from the tree at `/home/rog/odl20lite` and from runs executed on 15 September 2026. The competitive and valuation sections are judgement, not repository evidence.